

Bhagyesh Rathi

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EDUCATION

San Jose State University

Master of Science in Artificial Intelligence

San Jose, CA

Expected: May 2027

San Jose State University

Bachelor of Science in Computer Science, Magna Cum Laude

San Jose, CA

August 2020 - May 2024

RELEVANT EXPERIENCE

Rakuten - Software Engineer Co-op | San Mateo, CA

August 2023 – December 2023

- Built **event listeners** to process consent-revocation and account-deletion signals from Google, Facebook, and Apple OAuth providers.
- Validated and decoded **JWTs** for secure inter-service authentication; deserialized JSON payloads for downstream processing.
- Architected an internal **SDK** abstracting service-to-service communication, published to **Artifactory** via **GitLab CI/CD**.
- Achieved **85%** test coverage (**JaCoCo**) through unit and integration testing; eliminated code smells via **SonarQube** analysis.
- Created **Dockerfiles** and orchestrated containers using **Kubernetes**, deploying across non-prod and prod environments.
- Integrated **OpenTelemetry** and **Google cloud Trace** for real time **monitoring** and performance analysis.

Rakuten - Software Engineer Intern | San Mateo, CA

May 2023 – August 2023

- Contributed to **Social Sign Up / Login** with Google, Apple, Facebook for Rakuten businesses in Spain, UK, and Germany.
- Built a Kotlin microservice for social account linking — display and unlink flows for Google, Facebook, and Apple.
- Drove **40%** adoption of Social Authentication, lifting CVR **37%** over password-based auth.
- Designed REST APIs using OpenID Connect and OAuth 2.0 for token-based authentication and authorization.

PROJECTS

RAG-Powered Interactive Portfolio | Next.js, Pinecone, Vertex AI, Vercel AI SDK, Langchain | [GitHub](#), [Chat](#)

Feb 2026

- Shipped a production-grade **RAG** chatbot serving real users on a live portfolio site with full ingestion pipeline.
- PDF ingestion → recursive chunking → 768-dim embeddings → Pinecone upsert, with top-K retrieval, score thresholding, and metadata filtering to minimize hallucination
- Built a streaming API endpoint using **Vercel AI SDK** with **Gemini 2.5 Flash**, achieving sub-second TTFT.
- Implemented multi-layer security: input validation, system prompt hardening against **prompt injection**, and topic-scoped guardrails
- Authored a **27-test integration suite** covering **embedding quality**, **retrieval accuracy**, **pipeline correctness**, and **injection defense**.

SRE Copilot: AI Incident Triage Agent for On-Call SREs | Python, PyTorch, LangChain, ChromaDB | [Colab](#)

Feb 2025

- Built a **ReAct tool-use agent** that triages production incidents across 5 data sources (**RAG** over 768 runbooks, text-to-SQL, GCP status API, vulnerability DB, web search) using dual **Qwen 2.5** models (32B + 3B) on a single GPU via **4-bit quantization**.
- Evaluated **4 prompting techniques across 80 benchmarks**, finding self-reflection improved answer **grounding** by **125%** (0.40→0.90) and **actionability** by **100%** (0.50→1.0) over baseline, while prompt chaining achieved perfect tool selection on the 32B model but failed on 3B — revealing minimum capability thresholds for structured data.
- Achieved ~4% prefill speedup on the **32B** model by building **KV cache** reuse for **system prompt acceleration**.
- Implemented **defense-in-depth security** (SQL injection blocking, prompt hardening) passing 3/5 red-team attacks per model; deployed via Gradio with real-time model/technique comparison.

Caption Lens: Visual-Semantic Alignment via Attention Mechanisms | PyTorch, Hugging Face| [App / Repo](#)

May 2025

- Designed and trained three **encoder-decoder image captioning** models (Show-and-Tell, Show-Attend-and-Tell with soft attention, Visual Sentinel adaptive attention) on **MS COCO 2014 (82K images, ~9K-word vocab)** using a fine-tuned **ResNet-101** encoder feeding 7×7×2048 spatial features into an LSTM decoder.
- Implemented Bahdanau soft attention with doubly stochastic regularization, boosting **CIDEr** from **1.023** → **1.040** and winning all 8 metrics (**BLEU-1-4**, **METEOR**, **ROUGE-L**, **CIDEr**, **SPICE**); **BLEU-4** of 0.329 exceeds the original 2015 paper (0.250).
- Engineered beam search with 1-10 beam-width ablation lifting CIDEr from 1.013 (greedy) to 1.093 (beam=6); built interpretable attention heatmaps tracing each word to its driving image region.
- Deployed as an interactive multi-model demo on **Hugging Face Spaces** via **Docker** + **Gradio** with real-time model switching.

SKILLS & CERTIFICATIONS

Programming Languages: Java, Python, JavaScript, SQL

Frameworks: Flask, Scikit-Learn, Tensorflow, Pytorch, Pandas, Numpy, Seaborn

AI/ML Tools & Frameworks: Vertex AI, Google ADK, HuggingFace, LangChain, Ollama, Claude code, Gemini, Copilot, Cursor

Skills: Distributed Systems, Multi-tiered Systems, Agile software development, Unit/Integration Testing, Monitoring and Alerting.

Databases: MongoDB, MySQL, PostgreSQL, Redis

Cloud: GCP, Docker, Kubernetes, AWS, Vercel

Certifications: [Machine Learning Specialization\(Stanford\)](#)